registered. Although usually this is not as severe as a permanent failure, in mission critical applications such as when firing a missile with a warhead, or when guiding a spacecraft into orbit around a distant planet or moon, even a temporary short can be catastrophic.

Debris: Whiskers may grow as long as 10 mm, and although they might not even form in areas that have circuits (and thus aren't likely to cause shorts), they can still affect systems. For example, a whisker that's detached and finds its way into a precision telescope out in space will forever ruin the images that that telescope produces. With precision moving parts, such as with finely crafted watches, or other precision instruments with tiny gears, a metal whisker can jam up the system and effectively ruin it.

Plasma arc: As we mentioned earlier, a temporary short occurs when the current in the system is higher than the fuse current of the whisker. However, in some special cases, where the current flowing is orders of magnitude higher than the fuse current and the atmospheric conditions are just right, a whisker can fuse and go from solid to plasma state. Once in a plasma state, the conductivity of the arc is several orders of magnitude higher than a whisker is capable of. For example, if a tin whisker is capable of conducting 50 milliamperes before fusing, plasma arcs of tin have caused short circuits that can carry hundreds of amperes of current for several seconds until whatever circuit breaker is installed kicks in. This is a huge amount of current, and causes things to get very, very hot, very, very fast, and results in permanent damage of electronics as they burn up, and possibly even explode. A couple of satellites have had this happen in space, and have (presumably) had spectacular blowouts as the plasma arc caused the mother of all short circuits!

THEORIES

While we've become quite adept at minimising whisker formation using several methods that are too complex to get into and beyond the scope of this article, scientists are still left pondering why whiskers occur, and although we have no definitive explanation there are a couple of theories that have been put forth.

One of the theories we've discussed above, which was from the 1950s and showed that the greater the force applied to the layer, the more the metal will whisker. From this observation and others made over the years, a theory was born that when the pressure of tin atoms at the base of a film is higher than at the surface, and also when there is a pressure difference acting along the surface, a whisker is likely to form. The analogy the researchers used to explain this to us laymen was that of a toothpaste tube – no matter where you press (at the base, along the sides... anywhere) it causes toothpaste to come out of the tube. Thus, on the surface, the points of lowest pressure (of atoms) act like an opened tube of toothpaste, and a whisker sprouts. This theory arises primarily from research that was done by scientists at the Max Planck Institute of Metals Research along with researchers from Robert Bosch GmbH in 2009. However, it doesn't explain why some metals whisker more easily than others, despite similar force differences being recreated. Clearly, it's not just mechanical stresses and pressures that can cause whiskering, but some physical properties are also at play.

Metal whiskers are thought to have ended the Galaxy IV satellite mission. About 80% of pagers in the US stopped working for a day because this satellite died on May 19, 1998

More recently, in 2014, a physicist decided to challenge the idea of mechanical stresses, and published a theory that puts the spotlight on electrical currents instead. Victor Karpov from the University of Toledo, Ohio, has a theory that electrical charges might be the reason for whiskering. All along a metal's surface, defects and even mechanical stress can create positive and negative charges in patches. Basic physics tells us that like charges repel and opposites attract. Since the surface has rather large patches (relative to whiskers) of a single charge, this causes a strong electric field that pushes atoms with the same charge away (that is, upwards). While being perfectly reasonable, there are still questions left unanswered. One of the predictions that Karpov's makes is that whisker growth would be painfully slow to get started, but would eventually speed up. This has been checked and experimentally verified. His theory also explains why increasing stress on the metal will result in faster whisker growth – something we've known about since the 1950s but not yet explained. However, his theory doesn't explain why some metals are more prone to whiskering as compared to others, and also isn't able to describe the forces acting at the microscopic level.

Despite both the theories we've mentioned in passing (you should read the full papers) seeming plausible, the fact is that there are experimental observations that neither theory can address. Although we're not totally in the dark about metal whiskering, the answers haven't dawned on us just yet. In an age where one is usually marvelling at the stuff we know, these little whiskers remind us that we're still pretty ignorant of a lot more than what we know. But then again, isn't that ignorance exactly what makes science so fascinating?